

Problem 1

(a) Shown in class the basis for the product topology of $\mathbb{R}^\omega$ are of the form

$$U_1 \times U_2 \times \cdots \times U_n \times \mathbb{R} \times \mathbb{R} \times \cdots$$

s.t. U_i are open sets in $\mathbb{R}$. Let $x \in \mathbb{R}^\omega$. Let U be an open neighborhood of x . It suffices to show for just a basis set, so let U just be a basis element.

Define a sequence $y = (y_1, y_2, \dots)$ s.t.

$$y_i = \begin{cases} x_i, & \text{if } 1 \leq i \leq n, \\ 0, & \text{if } i > n. \end{cases}$$

Thus, $y \in U \cap \mathbb{R}^\infty$

Thus, every open neighborhood of every point $x \in \mathbb{R}^\omega$ intersects $\mathbb{R}^\infty$. Thus $\overline{\mathbb{R}^\infty} = \mathbb{R}^\omega$ in the product topology.

(b) Let $x = (x_1, x_2, \dots) \notin \mathbb{R}^\infty$ this means that there is no $n \in \mathbb{N}$ s.t. it is all zeros after. Thus consider the neighborhood of x

$$U = \prod_{i=0}^{\infty} U_i$$

s.t.

$$U_i = \begin{cases} (-1, 1), & \text{if } x_i = 0, \\ (x_i - \epsilon_i, x_i + \epsilon_i), & \text{if } x_i \neq 0. \end{cases}$$

$\epsilon_i = |x_i|/2$. This makes sure that the interval does not intersect zero.

This is open since it is made up of open intervals of $\mathbb{R}$.

Let $y \in U$. By definition of U , for each n we have $y_n \in U_n$.

If $x_n \neq 0$ then $y_n \neq 0$, and since there is no n where it ends it infinitely many zeros this means y can't also have infinitely many zeros. Thus $y \notin \mathbb{R}^\infty$

Thus, $U \cap \mathbb{R}^\infty = \emptyset$.

Thus, $\overline{\mathbb{R}^\infty} = \mathbb{R}^\omega$, since any point outside we can create a neighborhood that does not intersect $\mathbb{R}^\infty$

(c) Let $C = \{x = (x_1, x_2, \dots) \in \mathbb{R}^\omega \mid (x_n) \rightarrow 0\}$

Let $x \notin C$ this means x_n does not converge to 0. Thus, there is a $\epsilon_0 > 0$ s.t. for all

$N \in \mathbb{N}$ there is some $n \geq N$ with

$$d(x_n, 0) \geq \epsilon_0$$

Fix $\epsilon = \min\{1, \epsilon_0/2\}$. Consider the open ball

$$B(x, \epsilon) = \{y \in \mathbb{R}^\omega \mid d(x, y) < \epsilon\}$$

since $\epsilon < 1$ we get that

$$B(x, \epsilon) = \{y \in \mathbb{R}^\omega \mid d(x_n, y_n) < \epsilon \text{ for all } n \in \mathbb{N}\}$$

Assume for contradiction that $y \in B(x, \epsilon) \cap \mathbb{R}^\infty$

Fix $M \in \mathbb{N}$ s.t. $y_n = 0$ for all $n \geq M$.

However, since $x \notin C$ we have that $d(x_n, 0) \geq \epsilon_0$ for infinitely many $n \in \mathbb{N}$ thus pick index $n_0 \geq M$ s.t.

$$d(x_{n_0}, 0) \geq \epsilon_0$$

Then since $y_{n_0} = 0$ we have

$$d(x_{n_0}, y_{n_0}) = d(x_{n_0}, 0) \geq \epsilon_0 > \epsilon$$

Thus $y \notin B(x, \epsilon)$ which is a contradiction therefore $x \notin \overline{\mathbb{R}^\infty}$, since there is a neighborhood of x that does not intersect $\mathbb{R}^\infty$

Let $x \in C$ thus $(x_n) \rightarrow 0$

Let U be an open neighborhood of x . Thus there exists an $\epsilon > 0$ s.t.

$$B(x, \epsilon) \subseteq U$$

If $\epsilon > 1$ just fix $0 < \delta < 1$ s.t. $B(x, \delta) \subseteq B(x, \epsilon) \subseteq U$, so WLOG let $\epsilon < 1$.

Since $(x_n) \rightarrow 0$ there is an $N \in \mathbb{N}$ s.t. for all $n \geq N$ we have

$$d(x_n, 0) < \epsilon/2$$

Define a sequence $y = (y_1, y_2, \dots)$ s.t.

$$y_n = \begin{cases} x_n, & n < N, \\ 0, & n \geq N. \end{cases}$$

For $n < N$ we have that $y_n = x_n$ so $d(x_n, y_n) = 0 < \epsilon$

For $n \geq N$ we have that $y_n = 0$ and since $d(x_n, 0) < \epsilon/2$ we have that

$$d(x_n, y_n) = d(x_n, 0) < \epsilon/2 < \epsilon$$

Thus for all $n \in \mathbb{N}$ we have $d(x_n, y_n) < \epsilon$ so

$$d(x, y) = \sup_{n \in \mathbb{N}} d(x_n - y_n) < \epsilon$$

Thus $y \in B(x, \epsilon) \subseteq U$

Thus $x \in \overline{\mathbb{R}^\infty}$

Thus $\overline{\mathbb{R}^\infty} = C$

Problem 2

(a) Assume f_n converges uniformly to f . Let O be open in Y . Consider $f^{-1}(O)$.

Let $z \in f^{-1}(O)$. Thus $f(z) \in O$. Thus there exists some $\epsilon > 0$ s.t.

$$B(f(z), \epsilon) \subseteq O$$

By definition of uniform convergence there is an $N \in \mathbb{N}$ s.t. for $n \geq N$ we have

$$d(f_n(x), f(x)) < \epsilon/3$$

for all $x \in X$

Since $f_N(x)$ is continuous at z there exists an open neighborhood U of z s.t. for all $x \in U$

$$d(f_N(z), f_N(x)) < \epsilon/3$$

Thus, for every $x \in U$, we have

$$d(f(x), f(z)) \leq d(f(x), f_N(x)) + d(f_N(x), f_N(z)) + d(f_N(z), f(z)) < \epsilon/3 + \epsilon/3 + \epsilon/3 = \epsilon$$

Thus for every $x \in U$, we have

$$f(x) \in B(f(z), \epsilon) \subseteq O$$

Thus $U \subseteq f^{-1}(O)$

Thus for any $z \in f^{-1}(O)$ there exists an open neighborhood U in X s.t. $U \subseteq f^{-1}(O)$.

This shows that $f^{-1}(O)$ is open. Thus f is continuous.

(b) Classic Example: Let $f_n(x) = x^n$ on $[0, 1] \rightarrow \mathbb{R}$.

If $0 \leq x < 1$

If $x = 0$ then let O be an open neighborhood of 0. Thus there is an $\epsilon > 0$ s.t.

$$B(0, \epsilon) \subseteq O$$

Since $x^n \geq 0$ for all n on the interval $[0, 1]$ we get that $d(x^n, 0) < \epsilon$

If